

Modern Portfolio Theory

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Efficient frontier of risk vs. return

PROFESSIONAL DEFINITION

Modern Portfolio Theory (MPT) shows how combining assets with imperfectly correlated returns reduces portfolio risk for a given expected return, defining an efficient frontier of optimal portfolios.

WHO DEVELOPED IT

Developed by Harry Markowitz in his 1952 paper 'Portfolio Selection', for which he later received the Nobel Prize.

WHY IT WAS DEVELOPED

It was developed to formalise diversification, showing mathematically that risk depends on covariance, not just individual asset risk.

FIGURE 1 · MODERN PORTFOLIO THEORY

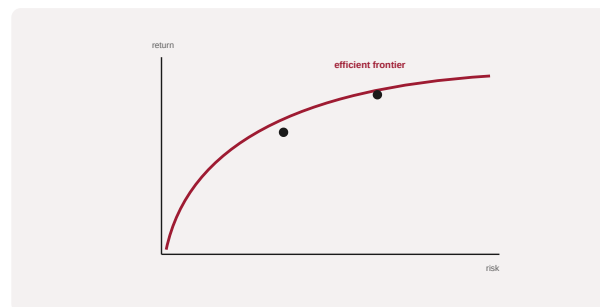


Figure 1. A schematic of the Modern Portfolio Theory as applied in BARBM312 (illustrative).

HOW IT IS USED

Assets are combined to minimise variance for a target return; portfolios are selected from the efficient frontier per the investor's risk appetite.

WHEN IT IS USED

When constructing portfolios and assessing diversification, including AI-driven allocation.

BY WHOM IT IS USED

Portfolio managers, investment analysts and advisers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It established diversification as the 'only free lunch' in investing and founded quantitative portfolio construction — a law AI accelerates but does not repeal.

RELATED FRAMEWORKS IN THIS COURSE

Risk-Return Spectrum · Slibrary 21

Diversification Curve · Slibrary 21

Capital Asset Pricing Model · Slibrary 21

Black-Litterman Model · Slibrary 21

SOURCES (HARVARD REFERENCING STYLE)

Markowitz, H. (1952) 'Portfolio selection', The Journal of Finance, 7(1), pp. 77–91.

Markowitz, H. (1959) Portfolio Selection: Efficient Diversification of Investments. New York: Wiley.